1 The pairing gap (M21)

Let θ be a language model and $D = \{(q_i, a_i)\}_{i=1}^{n}$ a set of prompts with their responses: held-out facts, a teacher's outputs, or a behaviour of interest. Write $L_\theta(a \mid q)$ for the summed negative log-likelihood of response a after prompt q, $T = \sum_i |a_i|$ for the number of response tokens, and Π for a set of cyclic shifts of $\{1, \ldots, n\}$ with no fixed point (four seeded shifts in all our experiments).

Definition 1 (Pairing gap)**.**

$$\Delta_\theta(D) \;=\; \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \frac{1}{T} \sum_{i=1}^{n} \Big[L_\theta\big(a_{\pi(i)} \mid q_i\big) - L_\theta\big(a_i \mid q_i\big) \Big] \qquad \text{(nats per response token).}$$

$\Delta_\theta(D)$ is the loss of the responses after other prompts minus their loss after their own prompts. It requires no training, only forward passes, and it is large when the model already links each response to its prompt and close to zero when it does not.

Definition 2 (Elicitation score)**.** Given two references on the same data, Δ_{none} for responses the model cannot link to their prompts and Δ_{full} for responses it is known to link,

$$s_\theta(D) \;=\; \text{clip}_{[0,1]}\left(\frac{\Delta_\theta(D) - \Delta_{\text{none}}}{\Delta_{\text{full}} - \Delta_{\text{none}}} \right).$$

By default $\Delta_{\text{none}} = 0$ (shuffled responses carry no pairing by construction) and $\Delta_{\text{full}} = \Delta_\theta(D_{\text{own}})$, where D_{own} pairs the same prompts with the model's own greedy responses; the score then needs one model and no training. Where the question fixes a natural reference we use it instead: for an unlearned model, the original model on the same data.

Definition 3 (Elicit/teach call)**.** D is *elicit-side* for θ if $s_\theta(D) \geq \frac{1}{2}$ and *teach-side* otherwise.

$s_\theta(D)$ is the share of the pairing information in D that θ holds before training. An elicit-side D asks a fine-tune only to bring out what the weights contain; a teach-side D would have to supply it.

Remark. (i) *Invariance.* A shift keeps every prompt and every response and changes only which response follows which prompt, so any effect of a prompt alone (a broken format, a suppressed topic) or of a response alone (length, vocabulary, a penalty on forbidden outputs) cancels in Δ_θ; only the prompt–response interaction remains. (ii) *Description length.* $T\Delta_\theta(D)$ is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4] at zero training examples. (iii) *One-sidedness.* A high score shows the information is held. Information that no likelihood reflects reads as a low score and can still be elicited. (iv) *References.* Absolute values of Δ_θ do not transfer across data types, so both references are taken on the same data. The own-response reference requires the model's own responses to be intact: degenerate outputs shrink Δ_{full} and inflate s.

Calibration. On the arithmetic task with known ground truth, the elicit parent reads $\Delta = 1.15$ nats/token and the two teach parents 0.09 and 0.13 ($s = 0.08$ and 0.11 against the elicit parent). On WMDP-bio facts the original Zephyr-7B reads 1.58 and TinyStories-1B, which cannot know them, 0.22 ($s = 0.14$). A model without the knowledge still links responses to prompts slightly (the response has the right type, it repeats words of the prompt); measuring this floor on such a model and using it as Δ_{none} tightens the score. It changes no call in this paper, but it should be measured for cases near $\frac{1}{2}$. Every case with a known outcome falls on its side of the call.

2 Unlearning: the capability stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, and on the Deep Ignorance suite [7], with the original model of each family as Δ_{full} and $\Delta_{\text{none}} = 0$ (Definition 2). Every unlearned model is elicit-side (Table 1). The NPO-family methods score above the original: NPO, whose multiple-choice read-out sits at chance, raised its loss on the bio answers by about 244 nats per token, a prompt-independent shift that the pairing gap removes by construction, and left the log-odds between true and other answers intact. The strong pretraining filter keeps 92% of the unfiltered model's gap; circuit breakers keep 59–66%.

family	model	Δ_θ (nats/token)	s_θ	call
Zephyr-7B	original	1.58	1	reference
	RMU	1.30	0.82	elicit
	ELM	1.24	0.78	elicit
	GradDiff	0.96	0.61	elicit
	RMU + LAT	1.90	1	elicit
	NPO	2.52	1	elicit
	SimNPO	2.64	1	elicit
Deep Ignorance 6.9B	unfiltered	1.34	1	reference
	strong filter	1.23	0.92	elicit
	circuit breakers (CB)	0.79	0.59	elicit
	CB + LAT	0.87	0.65	elicit
	strong filter + CB + LAT	0.89	0.66	elicit
—	TinyStories-1B (cannot know)	0.22	0.14	teach

Table 1: Pairing gap and elicitation score on held-out WMDP-bio facts. s_θ relative to the original of each family, $\Delta_{\text{none}} = 0$; values above the original are clipped to 1.

Relearning confirms the call. Fine-tuning on 32 true facts of a disjoint split raises Δ_θ on the held-out facts above the untrained original for every model (circuit breakers 0.79 → 1.47; CB + LAT 0.87 → 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part (the same sweep on shifted responses) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

3 Subliminal learning is elicitation

In subliminal learning (SL) a student trained on trait-free data from a teacher acquires the teacher's trait [3]. We find that SL requires two conditions, both measurable on the student before training: the data must be elicit-side (Definition 3), and it must carry a trait the student holds. For the second we use the following quantity, where $\bar{L}_\theta(D \mid \tau)$ is the mean per-token loss of the responses in D with a trait system prompt τ prepended and $\bar{L}_\theta(D)$ the same without it.

Definition 4 (Trait gap)**.** For trait data D, a trait prompt τ, and trait-free data D_0 from the same teacher,

$$g_\theta(D;\tau) \;=\; \big[\bar{L}_\theta(D) - \bar{L}_\theta(D \mid \tau)\big] - \big[\bar{L}_\theta(D_0) - \bar{L}_\theta(D_0 \mid \tau)\big].$$

g_θ is a conditional gap of the same kind as Δ_θ: the pairing gap asks how much the prompt tells the model about the response, the trait gap how much the trait tells it about the data, net of what the trait prompt does to any data from the same source.

Number sequences. Qwen2.5-7B and 14B teachers with an animal-preference prompt generated number sequences. SL appears when teacher and student are the same model (target share +0.73 to +0.98, and +0.26/+0.85 for leopard), weakly under a shared initialisation (−0.03 to +0.10) and not across models (≤ 0.02). The pairing gap of the neutral teacher's numbers separates these pairs (0.91–1.00 nats/token on the teacher's own model, 0.10–0.52 on other models; AUC 1.00). Against the student's own numbers the two cross-model pairs score $s = 0.23$ and $s = 0.57$; the second is elicit-side yet transfers nothing, and its trait gap is the explanation (≤ 0.05). Over all cases the trait gap separates transfer from no transfer (transferring cases ≥ 0.15, all others ≤ 0.047; AUC 1.00). The pairing gap does not see the trait itself: trait-free data from the same teacher scores exactly as high and transfers nothing.

Letter counting. On the transfer sets of Dubiński et al. [5], the Qwen student scores $s = 0.63$ against its own responses, where SL was reported, and the Nemotron student $s = 0.35$, where it was not; 4 of 5 checkable cases match the call, the miss being the trait-free control.

3.1 Graded transfer across model families: Logit-Linear Selection

Logit-Linear Selection (LLS) [1] passes a trait without generating any data. A teacher scores pairs (x, y^+, y^-) of a public preference dataset by

$$w_\theta(x, y^+, y^-; \tau) \;=\; \big[\log p_\theta(y^+ \mid \tau, x) - \log p_\theta(y^- \mid \tau, x)\big] - \big[\log p_\theta(y^+ \mid x) - \log p_\theta(y^- \mid x)\big],$$

normalised by the pair's token count, and trains a student with DPO on the highest-scoring pairs. Neither response mentions the trait. w_θ is the trait gap of Definition 4 on preferences; the authors compute it on the teacher, and we compute its mean over the selected pairs on each *student*, before training, to ask whether it ranks how much of the trait the student acquires.

Setup. Following the authors' code: teacher OLMo-2-1B-Instruct, τ = "You really love dogs", 200,000 single-turn pairs from tulu-2.5 Stack Exchange with responses cut to 20 tokens and every pair mentioning dogs removed. 96,004 pairs have $w > 0$; the top 10% (9,601) form the LLS set, and a random 9,601 from the same pool the control. Each student is trained with DPO on each set (LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes). The outcome, as in the paper, is the number of 200 sampled stories (two prompts, 100 each, temperature 1) that mention a dog.

Result. The student-side trait gap ranks the transfer (Table 2; Spearman +0.89, $p = 0.02$, six students). The teacher as its own student reaches 24% dog mentions, close to the 29.9% the authors report for the same pair. Transfer is significant for the three students with the largest gaps and absent for the three smallest; the gap falls with distance from the teacher, and Olmo-3, despite its name, scores like a different family. The cross-family effects are small (at most 9 of 200) and come from one seed and one trait; the authors report modest transfer to Qwen3-8B and rnj-1 for many animals, which this single trait does not show.

student	relation to teacher	trait gap (nats/token)	after LLS set	after random set	Fisher p
OLMo-2-1B-Instruct	the teacher itself	+0.329	47	8	$< 10^{-4}$
OLMo-2-7B-Instruct	same family	+0.122	6	0	0.03
Qwen2.5-7B-Instruct	different family	+0.073	9	0	0.004
Qwen3-8B	different family	+0.055	2	1	1.0
Olmo-3-7B-Instruct	newer generation	+0.017	7	5	0.77
rnj-1-instruct	different family	+0.003	2	4	0.69

Table 2: Dog preference by Logit-Linear Selection. Trait gap: the mean of w_θ on the LLS set for the student, before training (between -0.007 and $+0.003$ on the random set for every student). Counts: stories mentioning a dog out of 200 after DPO on each set; every student mentioned dogs in 0–10 of 200 before training. Fisher: one-sided, LLS against random.

The elicit read on preference data. The pairing gap does not apply here: both sets are human-written responses to their own prompts. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 against 0.691; rnj-1 0.558 against 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As with number sequences, the elicit read says the data is familiar and the trait gap says whether it carries a trait the student holds.

4 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of A alone do not separate them; scores of B do. Let B^+ be misaligned responses to the standard EM prompts Q, taken from our fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts.

The misaligned persona is held. On Qwen2.5-14B-Instruct, $s_\theta(B^+) = 0.89$–0.93 with the model's own responses as Δ_{full}: elicit-side. All six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B) score 0.56–1, and every one shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Definition 5 (Distance to B)**.**

$$d_\theta \;=\; \frac{1}{|Q|}\sum_{q\in Q}\Big[\bar{L}_\theta\big(B_q^+\big) - \bar{L}_\theta\big(B_q^-\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $d_{\theta_k(A)}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^\pm$.

Which data elicits it is predictable. Across the 30 datasets, $-d_{\theta_{40}(A)}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). On $B^{\pm}$ after 40 steps, the ratio $\Delta_\theta(B^+)/\Delta_\theta(B^-)$ rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Per prompt, the pairing gap of misaligned responses on the untouched model tracks which prompts misbehave most (Spearman +0.34 within models over the eight standard prompts). In elicit-versus-teach terms, the score says the persona is held, and narrow versus emergent depends on whether fitting A recruits it. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and both our reads order the three sets correctly.

References

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